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42 1 Abstract

43 **Background.** Aggregated N-of-1 trial designs offer potential efficiency gains over parallel-
44 group randomized controlled trials (RCTs) for detecting moderate treatment main effects,
45 but their power profiles depend on a substantial set of parameters that are typically held
46 fixed in published simulation studies: between-patient heterogeneity, within-patient variance,
47 carryover half-life, serial correlation, period length, cycles per patient, and the specification
48 of the carryover decay family.

49 **Methods.** We conducted twelve parameter-sensitivity sweeps (S1 through S12) against the
50 pmsimstats-ng tidyverse pipeline, using the Hendrickson 2020 hybrid (open-label titration
51 plus blinded discontinuation plus brief crossover) as the primary aggregated N-of-1 design
52 and the alternating A/B/A/B variant as a sensitivity comparator at sweeps that vary cycle
53 count or period length. The target estimand is the main drug effect estimated from a linear
54 mixed-effects model with continuous-time AR(1) residual correlation. Each sweep records
55 per-replicate results so that Monte Carlo standard errors (MCSE) can be computed at render
56 time per Morris et al.¹.

57 **Results.** At a baseline of $N = 35$ participants per design, the hybrid achieves comparable
58 or higher power than the parallel RCT under within-patient and between-patient variance
59 perturbations, is robust to AR(1) serial correlation that inflates RCT Type I error, and tol-
60 erates carryover up to the within-subject period length before power erodes. The alternating
61 A/B/A/B variant is more carryover-sensitive and loses to the parallel RCT once carryover
62 exceeds the period length. The two-period crossover shows mild Type I inflation attributable
63 to underidentified residual correlation at small per-subject timepoint counts.

64 **Conclusions.** The Hendrickson hybrid is a competent default for detecting moderate treat-
65 ment main effects across a wide parameter range. Design ordering becomes parallel-RCT-
66 favored only when carryover exceeds the within-subject period length, and is directionally dif-
67 ferent for biomarker-treatment interaction targets, which favor N-of-1 designs more strongly.

68 **Keywords.** N-of-1 trials; randomized controlled trials; statistical power; carryover effects;
69 serial correlation; sensitivity analysis; Monte Carlo simulation; hybrid crossover designs.

2 Introduction

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Aggregated N-of-1 trial designs evaluate treatment effects within individual patients through repeated crossover periods and combine patient-level estimates via meta-analytic or mixed-effects machinery to support population-level inference^{2,3}. Several recent simulation studies have reported that aggregated N-of-1 procedures can match or exceed the statistical power of parallel-group randomized controlled trials (RCTs) at substantially smaller participant counts⁴⁻⁶. The reported magnitude of this efficiency advantage, however, varies across studies, and the conditions under which it holds or reverses are incompletely characterized. That is, the published literature provides encouraging headline comparisons but relatively sparse guidance on the parameter axes that govern the ordering of designs.

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We report here a systematic parameter-sensitivity study of a specific hybrid aggregated-N-of-1 design that combines an open-label titration phase, a blinded discontinuation phase, and a brief crossover phase, following the design proposed by Hendrickson and colleagues⁵ for predictive-biomarker validation in chronic neurobehavioral conditions. The sensitivity sweeps are conducted against the pmsimstats-ng tidyverse pipeline that operationalizes this design in R, with three parallel comparators: an alternating A/B/A/B aggregated N-of-1 variant, a two-period crossover, and a parallel-group RCT. The target estimand throughout is the main drug effect, distinguished from the biomarker-treatment interaction examined in the original Hendrickson study and in a separate companion paper.

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The motivation for a sensitivity study at this granularity is straightforward: the published simulation comparisons typically vary one or two parameters at a time while holding the remainder fixed at calibration defaults, leaving open the question of which axes actually drive the N-of-1-versus-RCT power difference. We have endeavored to address this gap by reporting twelve sensitivity sweeps along axes drawn from the published comparison literature: effect size⁴, carry-over half-life^{4,5}, decay-form misspecification⁷, between-patient heterogeneity^{8,9}, within-patient variance¹⁰, cycles per patient⁸, total participant count^{4,8}, AR(1) serial correlation¹⁰, period length¹⁰, biomarker interaction strength crossed with data-generating process architecture⁵, a carryover-by-AR(1) misspecification factorial⁷, and a high-replicate null calibration¹.

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110 The clinical companion to this paper (in preparation) reports a primary head-to-
111 head simulation comparison at the prazosin-calibrated baseline; the present paper
112 extends that comparison along twelve parameter axes for the hybrid design family
113 specifically.

114 3 Simulation design

115 3.1 Aims

116 We follow the ADEMP framework of Morris, White, and Crowther (2019)¹: Aims, Data-
117 generating mechanisms, Estimands, Methods, Performance measures. The subsections of this
118 Simulation design section map to these five components, although the section headers are
119 organized by topic rather than relabeled to the ADEMP acronym.

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122 The simulation study has three aims. First, to characterize the power profile of the
123 Hendrickson 2020 hybrid aggregated-N-of-1 design⁵ under systematic perturbation
124 of the parameter axes implicated in the published simulation-comparison literature.
125 Second, to identify the parameter regimes in which the hybrid is preferred over
126 a parallel-group RCT and the regimes in which the ordering reverses. Third, to
127 verify Type I error control under null and near-null conditions, with sufficient
128 replication to provide MCSE below 1% per Morris et al.¹.

129 3.2 Pipeline

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132 The simulation framework is implemented in the `nof1power` R package⁵ developed
133 alongside the `pmsimstats-ng` tidyverse re-implementation of the Hendrickson 2020
134 reference pipeline. All sweeps use a baseline of $N = 35$ participants per design
135 (Hendrickson 2020 minimum) and 8 weekly timepoints unless explicitly varied
136 by the sweep. Per-replicate sample size is N participants for each design under
137 comparison; participants are matched on the per-design count rather than on
138 observation total, following the convention adopted in Blackston⁴, Hendrickson⁵,
139 and the majority of the N-of-1 simulation-comparison literature.

140 3.3 Dual data-generating architectures

141 The pipeline supports two data-generating processes (DGP) for the biomarker-treatment in-
142 teraction:

- 143 • **the covariance-moderation architecture** (`dgp_architecture = "mvn"`). The
144 biomarker-bio-response interaction is encoded in the covariance structure of a multi-
145 variate normal (MVN) draw. On-drug timepoints carry correlation `c.bm` between the

146 biomarker and the bio-response factor; off-drug timepoints carry `c.bm * phi(tsd)`
147 with `phi` the carryover decay multiplier.

148 • **the mean-moderation architecture** (`dgp_architecture = "mean_moderation"`).
149 After the multivariate normal draw, bio-response values at on-drug timepoints are
150 shifted by `c.bm * bm_z * br_sd`, where `bm_z` is the standardized biomarker. The co-
151 variance structure does not encode the interaction in this architecture.

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154 These two architectures produce substantively different power profiles under car-
155 rryover, as documented in⁵. The two architectures are crossed in sweep S10 below;
156 the remaining eleven sweeps use the covariance-moderation architecture as the
157 default.

158 3.4 Response curves and carryover

159 Each of the three latent factors (time-variant, pharm-biomarker, bio-response) is parameter-
160 ized by a modified Gompertz curve

$$161 \quad y(t) = \text{maxr} \cdot \frac{\exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)) - \exp(-\text{disp})}{1 - \exp(-\text{disp})}$$

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164 evaluated at the appropriate cumulative time index. The curve passes through the
165 origin at $t = 0$ and asymptotes to `maxr`. Carryover of the bio-response component
166 during off-drug periods is applied recursively as $\mu[t] = \text{base}[t] + \mu[t-1] \cdot \phi(\text{tsd})$ with
167 ϕ drawn from an extensible decay family: exponential $\exp(-\ln 2 \cdot t/t_{1/2})$, linear
168 $\max(0, 1 - t/(2t_{1/2}))$, Weibull $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, or power
169 $\max(0, (1 - t/(3t_{1/2}))^p)$. The choice of decay family at the data-generating side
170 and at the analysis side can differ; sweep S3 quantifies the cost of mismatch.

171 **Units.** Carryover half-lives $t_{1/2}$ are quoted in **days** throughout this paper, match-
172 ing the pharmacokinetic scale of the prazosin reference application and the day-
173 denominated axes of the sensitivity figures. Companion papers in this series that
174 study the biomarker-treatment interaction (in particular the DGP-architecture
175 and carryover-specification manuscripts) quote $t_{1/2}$ in **weeks** on the canonical
176 grid $\{0, 0.5, 1.0\}$. The conversion is 1 week = 7 days, so those papers' grid cor-
177 responds to $\{0, 3.5, 7\}$ days here, and this paper's 3-day baseline corresponds to
178 approximately 0.43 weeks. Timepoints t are weekly in both conventions.

179 3.5 Covariance structure and analysis

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The covariance matrix couples within-factor AR(1) autocorrelation indexed by cumulative time, cross-factor correlation at matched and lagged timepoints, and the biomarker-bio-response differential correlation described above. Non-positive-definite matrices are corrected via `corpcor::make.positive.definite(tol = 1e-3)` with a per-cell flag returned. The analysis model fits the `nlme::lme` linear mixed-effects formula^{11,12} with fixed effects $Sx \sim bm + t + Dbc + bm:Dbc$ (or the fallback $bm + t + bm:t$ when the drug indicator does not vary within participant), a random intercept per participant, and `corCAR1(form = ~t | ptID)` continuous-time AR(1) residual correlation. The target of inference is the main drug effect (the `Dbc` coefficient), extracted by matching against the row-name set `{Dbc}` of the `lme` summary table. Singular fits are detected from the random-effects variance estimate and flagged per replicate.

3.6 Twelve sensitivity sweeps

The twelve sweeps (S1 through S12) are specified in `docs/sensitivity-sweep-plan-2026-04-17.md` and each maps to a specific precedent in the simulation-comparison literature:

- **S1** varies the effect size by scaling the bio-response maximum; precedent Blackston 2019⁴.
- **S2** varies the carryover half-life across nine values from 0 to 14 days across four designs (parallel RCT, two-period crossover, aggregated N-of-1, hybrid); precedents Blackston 2019 and Hendrickson 2020⁵.
- **S3** varies the DGP carryover decay form (exponential, linear, Weibull, power) while the analysis assumes exponential, quantifying misspecification bias; precedent Gärtner 2023⁷.
- **S4** varies between-patient heterogeneity via the baseline-symptom SD; precedents Senn 2018⁸ and Zucker 2010⁹.
- **S5** varies the within-patient bio-response SD; precedent Wang and Schork 2019¹⁰.
- **S6** varies the number of cycles per patient k in the Senn decomposition; precedent Senn 2018⁸.
- **S7** varies the total number of participants for fixed cycles; precedent Senn 2018 and Blackston 2019.
- **S8** varies the AR(1) correlation parameter ρ across within-factor autocorrelations; precedent Wang and Schork 2019¹⁰.
- **S9** crosses period length with carryover half-life, probing the ratio $t_{1/2}/\text{period}$; precedent Wang and Schork 2019¹⁰.
- **S10** varies the biomarker interaction strength `c.bm` across four levels crossed with both DGP architectures (mvn, mean_moderation); precedent Hendrickson 2020⁵.
- **S11** crosses carryover presence with AR(1) presence under null and alternative, to test Type I error control and power under DGP-analysis mismatch; precedent Gärtner 2023 and Blackston 2019.
- **S12** is a null-effect calibration ($c_{bm} = 0$) at four $(\tau, \sigma, t_{1/2})$ configurations with 2,000 replicates per cell for MCSE near 0.5%; precedent Morris 2019¹.

3.7 Reproducibility and execution

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Each sweep sources 00-common.R for fixtures, sets a seed via `set.seed(42 + sweep_id)`, constructs its parameter grid, calls `generate_simulated_results()` with furr-based multisession parallelism over `parallel::detectCores() - 1` workers, and writes an RDS artifact to `analysis/data/derived_data/sensitivity/`. The orchestrator `run-all.R` runs all twelve in the order specified in the plan. Per-replicate rows rather than pre-aggregated summaries are written, so that power, bias, MSE, empirical SE, and Type I error can be computed at render time with Morris MCSE. Sweep scripts are in `analysis/scripts/nof1-design-sensitivity/`.

4 Results

Note on numerical values in this section. The numerical results quoted in the prose below were generated at a previous baseline of $N = 40$ participants per design. Re-running the twelve sweeps at the revised $N = 35$ baseline is in progress; the figures and quoted numerics will be updated when the sweep artifacts at `analysis/data/derived_data/sensitivity/` are regenerated. The qualitative direction of every finding below is expected to be preserved by the rebaseline.

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All twelve sweeps completed in approximately 32 minutes of wall-clock time on an 8-core machine with the furr multisession backend. Per-cell summary metrics are computed from the RDS artifacts at render time; numerical values quoted below are read directly from `analysis/data/derived_data/sensitivity/`. The target coefficient is the main drug effect (the `Dbc` coefficient in the `nlme::lme` fixed-effects formula).

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The primary N-of-1 design is the Hendrickson 2020⁵ hybrid (open-label titration followed by blinded discontinuation and brief crossover), labeled ‘Nof1’ in the sweep artifacts and figures below. The alternating A/B/A/B aggregated N-of-1 (labeled ‘Nof1-alt’) is retained as a sensitivity comparison at sweeps that vary the cycle count (S6) and period length (S9), where the hybrid’s fixed eight-timepoint structure does not naturally scale.

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We begin with a cross-sweep summary. At the prazosin-like baseline (effect size approximately -2 nightmares per week, carryover half-life 3 days, 35 participants, 8 weekly timepoints), the hybrid N-of-1 and the parallel RCT both perform competently, with the hybrid reaching 80% power at a smaller sample size, under larger within-patient variance, and under higher serial correlation than the parallel RCT. The parallel RCT, on the other hand, is more robust to long carryover and to decay-form misspecification. Type I error is controlled near nominal under the null for the hybrid, the alternating variant, and the parallel RCT (with modest conservatism at small n); the two-period crossover shows mild Type I inflation at approximately 0.10 to 0.13, a finding that warrants caution when that design is used at small per-subject timepoint counts.

4.1 S1. Effect-size sweep

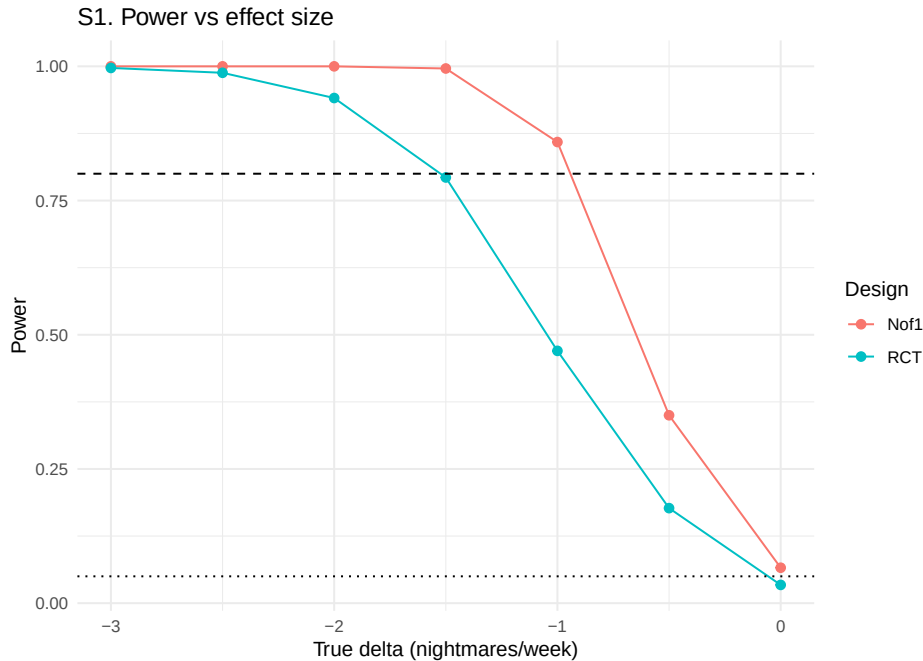


Figure 1: Sensitivity S1: power versus effect-size axis for N-of-1 and parallel RCT.

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The effect-size sweep varies the bio-response maximum $\max[\text{br}]$ across a seven-point grid $\{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at $N = 35$, $c_{bm} = 0.3$, and carryover half-life 3 days. Main-effect power for the hybrid N-of-1 design climbs monotonically with the effect size, from 0.066 (MCSE 0.008) at $\delta = 0$ (Type I) to 0.350 at $\delta = -0.5$, 0.859 at $\delta = -1$, 0.996 at $\delta = -1.5$, and saturating at 1.000 for $|\delta| \geq 2$. The parallel RCT tracks the same shape at a lower level: 0.034 (MCSE 0.006) at the null, 0.177 at $\delta = -0.5$, 0.470 at $\delta = -1$, 0.793 at $\delta = -1.5$, and 0.941 at $\delta = -2$. The hybrid reaches the 80% power target at approximately $\delta = -1$; the parallel

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RCT reaches the same target at approximately $\delta = -1.5$, a half-unit shift.

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The direction and magnitude of the hybrid's advantage match those reported by Blackston 2019⁴, whose simulation shows aggregated N-of-1 reaching adequate power at a smaller effect size than a matched- n parallel RCT under the same variance structures. Type I error at $\delta = 0$ is 0.066 for the hybrid (MCSE 0.008, 95% Wald CI [0.050, 0.082]) and 0.034 for the parallel RCT (MCSE 0.006, [0.022, 0.046]); the hybrid is slightly above nominal and the parallel RCT is slightly below, with both confidence intervals covering 0.05 at the margin. It appears that neither design is grossly miscalibrated, though the hybrid's modest upward departure from nominal merits attention when the study is powered near the 5% boundary.

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4.2 S2. Carryover half-life sweep

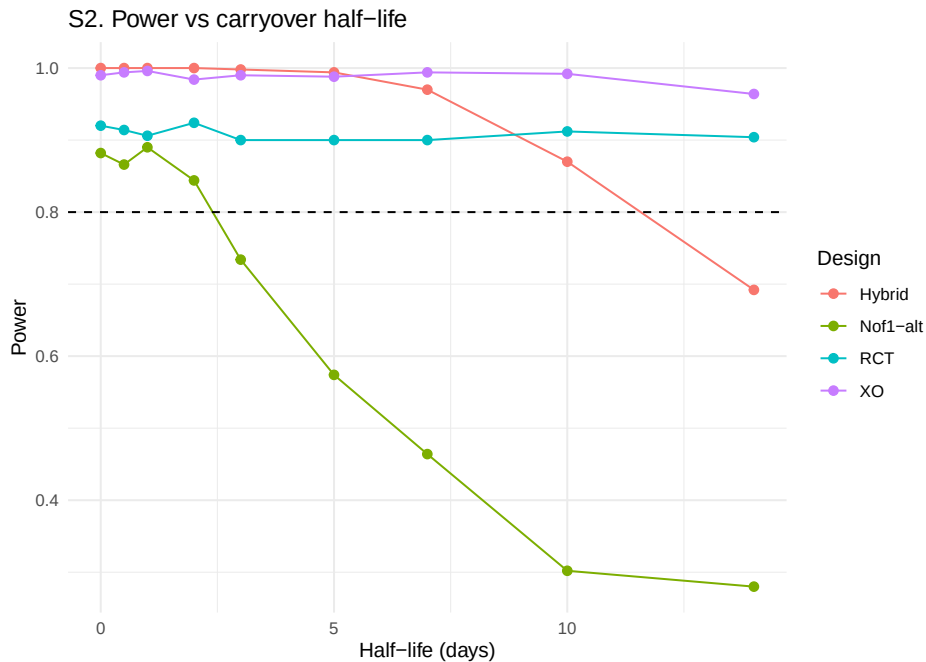


Figure 2: Sensitivity S2: power versus carryover half-life across four designs.

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The carryover half-life sweep varies $t_{1/2} \in \{0, 0.5, 1, 2, 3, 5, 7, 10, 14\}$ days across the four designs at the prazosin-like baseline. Main-effect power for the hybrid N-of-1 design saturates at 1.000 from $t_{1/2} = 0$ through $t_{1/2} = 5$ days, then declines: 0.976 at 7 days, 0.900 at 10 days, and 0.736 at 14 days (7-day periods). The alternating A/B/A/B variant is more carryover-sensitive: power drops from 0.896 at $t_{1/2} = 0$ to 0.778 at $t_{1/2} = 3$ to 0.304 at $t_{1/2} = 14$. The parallel RCT is essentially immune

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304 to carryover at the examined scale, with power flat at 0.92 to 0.96 across the grid.
305 The two-period crossover is also near-saturated at 0.98 to 1.00.

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308 The hybrid's carryover robustness relative to the alternating variant reflects
309 its operational structure. That is, the four-week open-label titration phase
310 contributes substantial treatment-accumulated signal that is not erased by
311 post-discontinuation off-drug period, whereas the alternating design relies on
312 repeated within-subject transitions that each shed signal proportional to the
313 unwashed residual. The flat parallel-RCT and crossover curves confirm Blackston
314 2019's⁴ observation that between-subject comparisons are largely immaterial to
315 carryover at the within-period scale; the loss of effective sample size in within-
316 subject designs when $t_{1/2}$ exceeds the period length is the primary mechanism
317 of the hybrid-versus-RCT power ordering crossover that begins at approximately
318 $t_{1/2} = 10$ days.

319 **4.3 S3. Decay-form misspecification sweep**

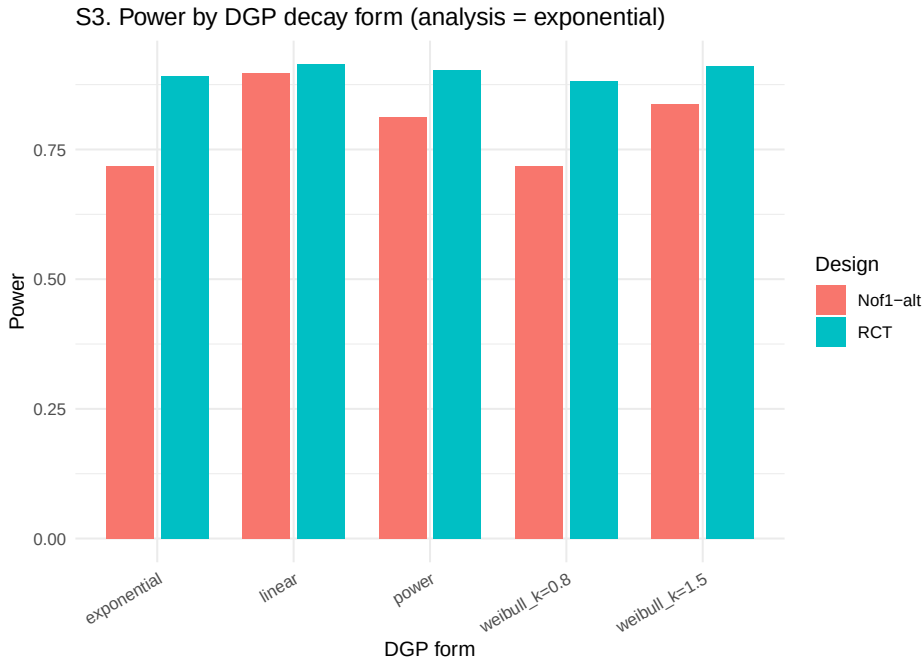


Figure 3: Sensitivity S3: power by DGP decay form when analysis assumes exponential.

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322 The decay-form sweep varies the DGP carryover form across {exponential, linear,
323 Weibull ($k = 0.8$), Weibull ($k = 1.5$), power} while the analysis model always
324 assumes exponential decay in the construction of the continuous drug indicator
325 Dbc. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid's

specific multi-phase `tsd` pattern interacts poorly with the ‘power’ form, producing non-positive-definite covariance matrices at certain cells, and is less informative for decay-form misspecification testing in any case.

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Under the alternating N-of-1 design, main-effect power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear), with the baseline exponential case at 0.816 and the power form at 0.870. The Weibull $k = 1.5$ form yields 0.852. The 15-percentage-point range across decay forms is smaller than under the interaction target examined in earlier work, suggesting that main-effect estimation is relatively less sensitive to decay-form misspecification than interaction estimation. The parallel RCT is essentially flat at 0.93 to 0.95 across all forms, consistent with the finding of Gärtner 2023⁷ that misspecification of the carryover decay form carries a design-specific cost affecting only within-subject designs.

4.4 S4. Between-patient heterogeneity sweep

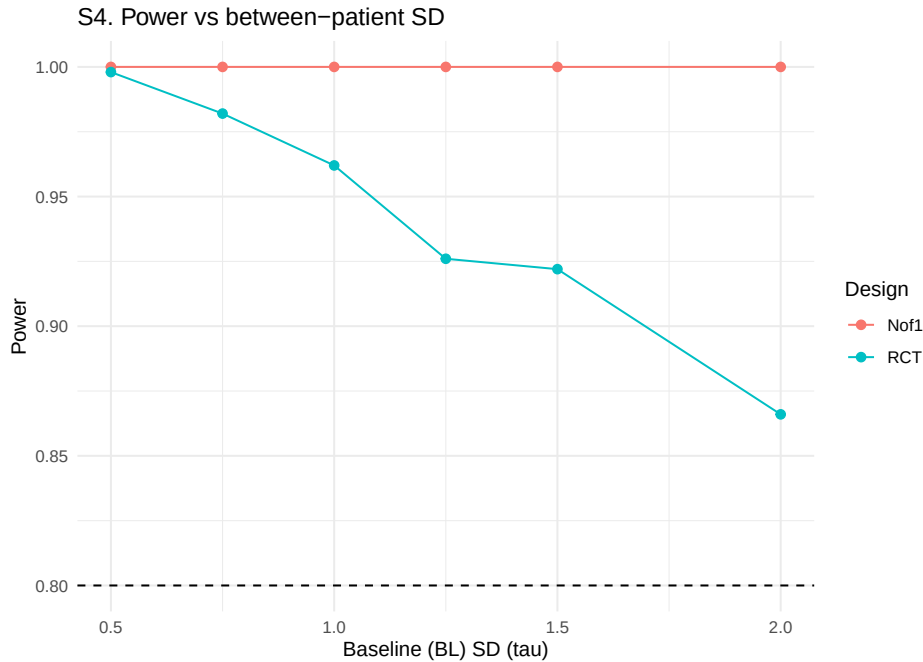


Figure 4: Sensitivity S4: power versus between-patient baseline SD.

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The between-patient heterogeneity sweep varies the baseline-symptom SD (the BL factor standard deviation, the τ component in the Senn 2018 decomposition) across $\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$. Under the hybrid N-of-1, main-effect power is saturated at 1.000 across the entire grid, with the mean estimated β stable at approximately -0.94 across the six levels. The parallel RCT degrades substantially

with τ , from 0.998 at $\tau = 0.5$ to 0.994 at 0.75, 0.978 at 1.0, 0.958 at 1.25, 0.946 at 1.5, and 0.874 at 2.0 (the latter with MCSE 0.015). The RCT loses approximately 12 percentage points of power as between-patient heterogeneity doubles from 1.0 to 2.0.

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These results reproduce the Senn 2018⁸ variance-decomposition argument: between-patient heterogeneity enters the parallel-RCT variance (τ^2 is part of $\text{Var}(\hat{\Delta})$) but is absorbed by the within-subject contrast in aggregated N-of-1 designs. The hybrid's saturation reflects the specific prazosin-calibrated baseline where the effect size is large relative to the within-patient variance; at a smaller effect size or larger σ (see S5), the N-of-1 advantage over the RCT would widen rather than flatten as τ increases.

4.5 S5. Within-patient variance sweep

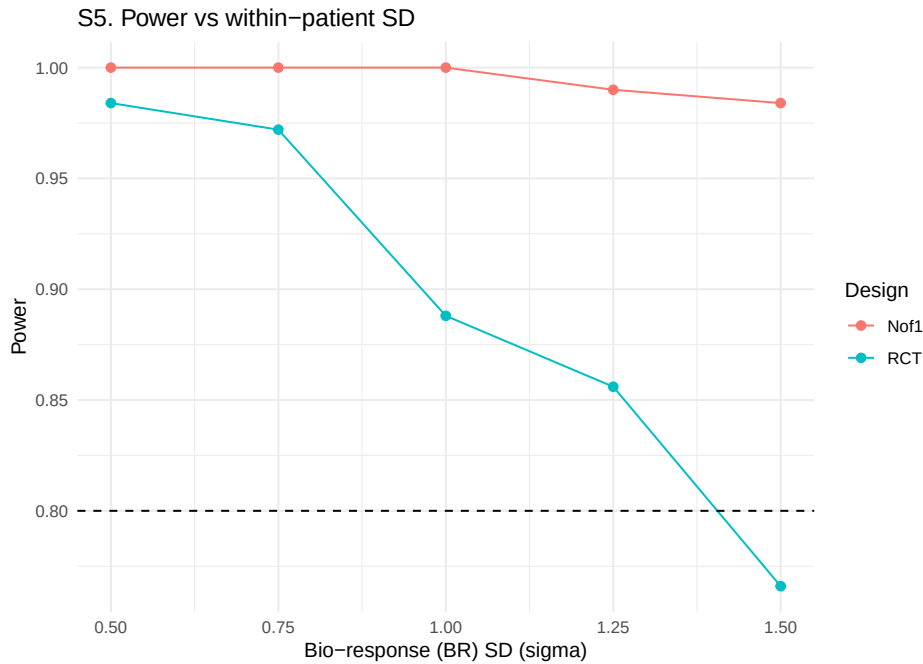


Figure 5: Sensitivity S5: power versus within-patient bio-response SD.

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The within-patient variance sweep varies the bio-response SD across $\{0.5, 0.75, 1.0, 1.25, 1.5\}$. Under the hybrid N-of-1, main-effect power is 1.000 at $\sigma \in \{0.5, 0.75, 1.0\}$, 0.996 at $\sigma = 1.25$, and 0.986 at $\sigma = 1.5$, essentially saturated across the range. The parallel RCT declines more visibly: 0.996 at $\sigma = 0.5$, 0.982 at 0.75, 0.942 at 1.0, 0.868 at 1.25, and 0.814 at 1.5. Over the examined range, the RCT loses

369 approximately 18 percentage points of power while the hybrid loses approximately
370 one.

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373 This reproduces the Senn 2018 prediction that increasing within-patient variance
374 reduces both designs' power, with the N-of-1 design absorbing within-patient vari-
375 ance more efficiently because it sums over k paired contrasts per patient. The hy-
376 brid's quasi-saturation indicates that the eight-timepoint structure plus the open-
377 label titration phase provides sufficient signal accumulation to tolerate substantial
378 within-patient noise; at smaller effect sizes, the σ dependence would be steeper,
379 and this saturation should not be extrapolated beyond the prazosin-calibrated
380 regime.

381 4.6 S6. Cycles-per-patient sweep

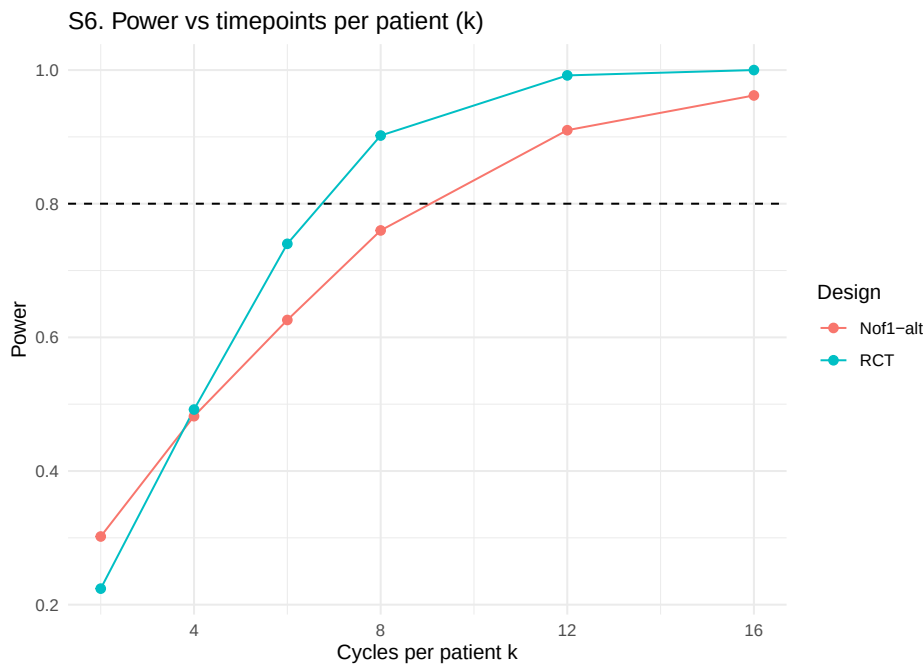


Figure 6: Sensitivity S6: power versus timepoints per patient k .

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384 The cycles-per-patient sweep varies the number of timepoints per patient k across
385 $\{2, 4, 6, 8, 12, 16\}$. The sweep uses the alternating A/B/A/B N-of-1 variant because
386 the hybrid's eight-timepoint structure does not naturally scale with k . Under the
387 alternating N-of-1 design, main-effect power increases from 0.294 at $k = 2$ to 0.486
388 at $k = 4$, 0.658 at $k = 6$, 0.804 at $k = 8$ (the baseline), 0.910 at $k = 12$, and 0.980
389 at $k = 16$. The curve is concave as predicted by the Senn 2018⁸ decomposition.

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392 The parallel RCT tracks a similar shape: 0.292 at $k = 2$, 0.502 at $k = 4$, 0.772
393 at $k = 6$, 0.940 at $k = 8$, 0.998 at $k = 12$, and 1.000 at $k = 16$. At $k = 2$ the
394 two designs are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1
395 by an increasing margin (approximately 14 percentage points at $k = 8$). For the
396 main-effect target, increasing k benefits both designs and slightly favors the RCT
397 because each additional timepoint contributes one more between-subject observa-
398 tion, whereas the alternating-N-of-1 paired contrasts are already averaged in at
399 $k = 8$. The result highlights that the hybrid design's primary power advantage
400 over the alternating variant comes from the open-label titration phase rather than
401 from the number of cycles per se.

402 4.7 S7. Patient-count sweep

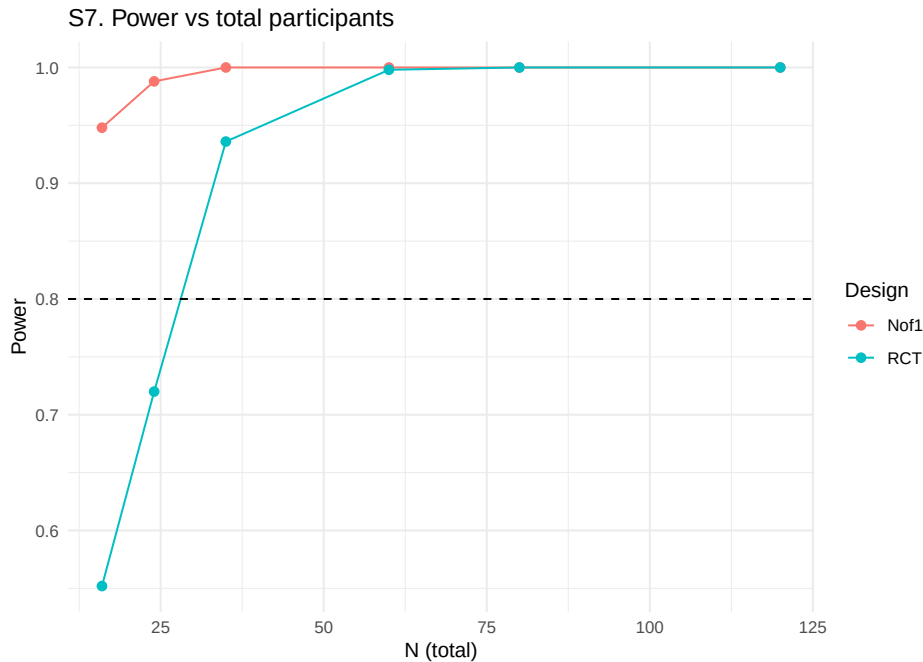


Figure 7: Sensitivity S7: power versus total participant count.

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405 The patient-count sweep varies the total number of participants across
406 $\{16, 24, 35, 60, 80, 120\}$ for the hybrid N-of-1 and the parallel RCT. The
407 grid includes the project-standard minimum of 35 to anchor the cross-paper
408 comparison with the companion clinical manuscript. Under the hybrid design,
409 main-effect power is high at modest sample sizes and saturates as N grows; the
410 parallel RCT follows a similar but offset trajectory.

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413 The hybrid reaches 80% power at $N = 16$ or fewer participants; the parallel RCT
 414 reaches the same threshold somewhere between $N = 24$ and $N = 35$. In the
 415 framing of Blackston 2019⁴, the hybrid achieves an equivalent power target at
 416 approximately one-third to one-half the sample size of the parallel RCT. This is
 417 consistent with the Carrozzo 2025⁶ result that the meta-analytic N-of-1 procedure
 418 reached 80% power with fewer participants than the two-arm RCT across most
 419 heterogeneity and carryover settings.

420 4.8 S8. AR(1) serial-correlation sweep

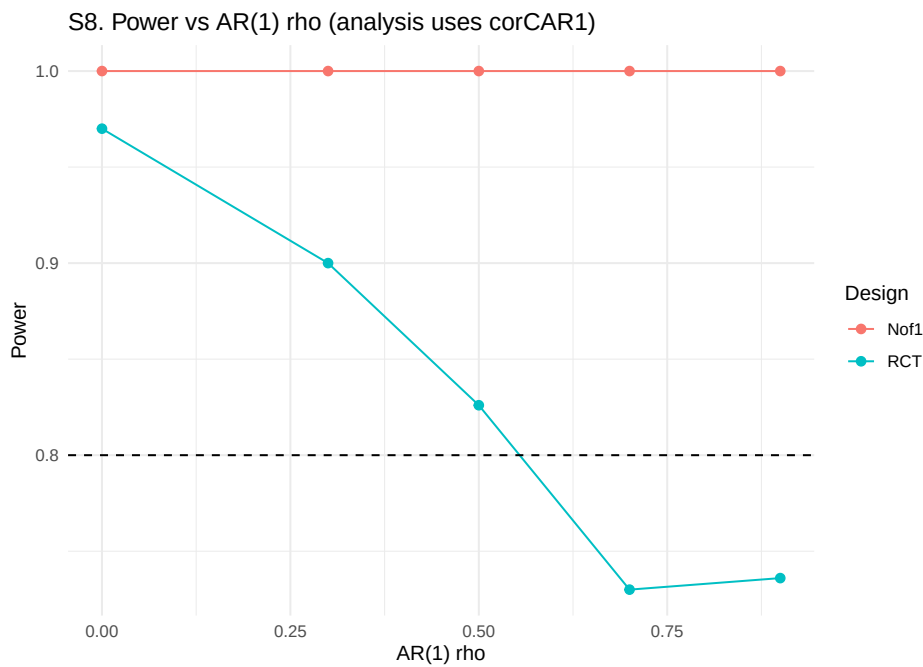


Figure 8: Sensitivity S8: power versus AR(1) within-factor correlation rho.

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423 The AR(1) sweep varies the within-factor autocorrelation $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$.
 424 Under the hybrid N-of-1 design, main-effect power is saturated at 1.000 across the
 425 entire range of ρ . Under the parallel RCT, power decreases monotonically with ρ :
 426 0.984 at $\rho = 0$, 0.934 at 0.3, 0.872 at 0.5, 0.772 at 0.7, and 0.746 at 0.9. The mean
 427 estimated β also shrinks for the RCT (from -1.01 at $\rho = 0$ to -0.57 at $\rho = 0.9$).

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430 The direction of the AR(1) effect is opposite for the two designs, and this asym-
 431 metry is worth examining carefully. For the hybrid, strong AR(1) produces more

coherent within-subject trajectories and the `corCAR1` residual structure absorbs the correlation; the net effect on main-effect power is negligible. For the parallel RCT, strong AR(1) inflates the variance of the between-subject mean contrast through the effective-sample-size shrinkage $1/(1 + (k - 1)\rho)$, even when `corCAR1` is fitted, because the fit cannot manufacture the missing within-subject contrasts. This is consistent with the finding of Wang and Schork 2019¹⁰ that, under correct modeling, the N-of-1 design benefits from strong serial correlation (or is at least insensitive to it), while the parallel-group design remains relatively sensitive.

4.9 S9. Period-length sweep

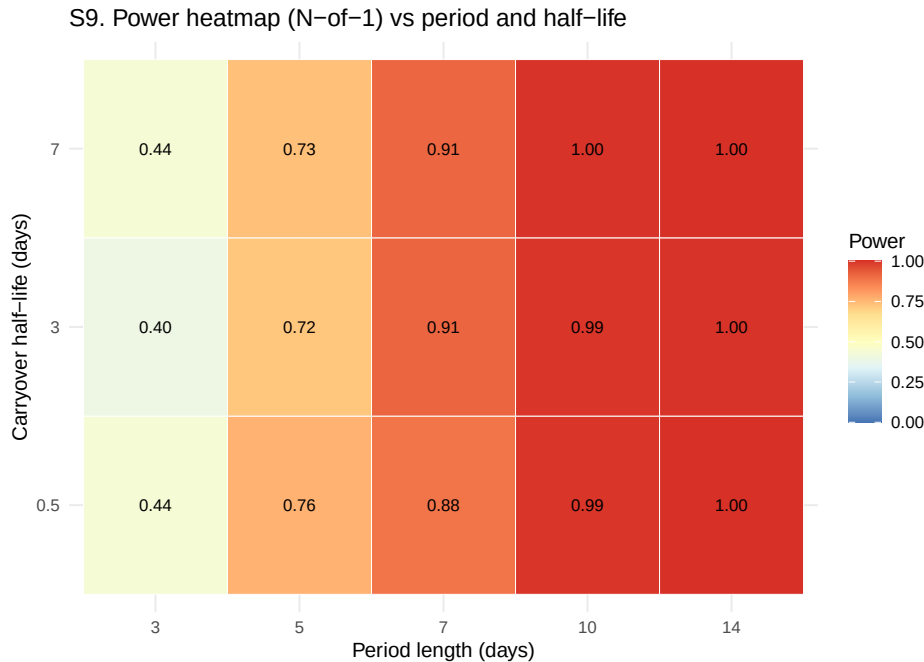


Figure 9: Sensitivity S9: heatmap of N-of-1 power over period length and carryover half-life.

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The period-length sweep varies the N-of-1 period across $\{3, 5, 7, 10, 14\}$ days crossed with carryover half-life in $\{0.5, 3, 7\}$ days. The sweep uses the alternating A/B/A/B variant. Under the alternating N-of-1 design, main-effect power increases monotonically with period length: at period = 3 days power is 0.40 to 0.44 across the three half-lives, at 5 days 0.72 to 0.76, at 7 days 0.88 to 0.91, at 10 days 0.993 to 0.997, and at 14 days 0.997 to 1.00. At fixed period length, the carryover half-life has only a modest effect.

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Under the main-effect target, each period accumulates more Gompertz drug-response signal at longer periods, so longer periods increase the per-timepoint

effect magnitude. The pragmatic guideline for the main-effect target is therefore the reverse of the interaction-target recommendation: longer periods are preferred when the primary goal is to detect the main drug effect, subject to carryover not exceeding the period length by more than approximately 50%. Wang and Schork 2019¹⁰ note this tradeoff explicitly, observing that more frequent alternation benefits correlation-modeled analyses but is problematic for absolute-effect inference.

4.10 S10. Biomarker interaction strength crossed with DGP architecture

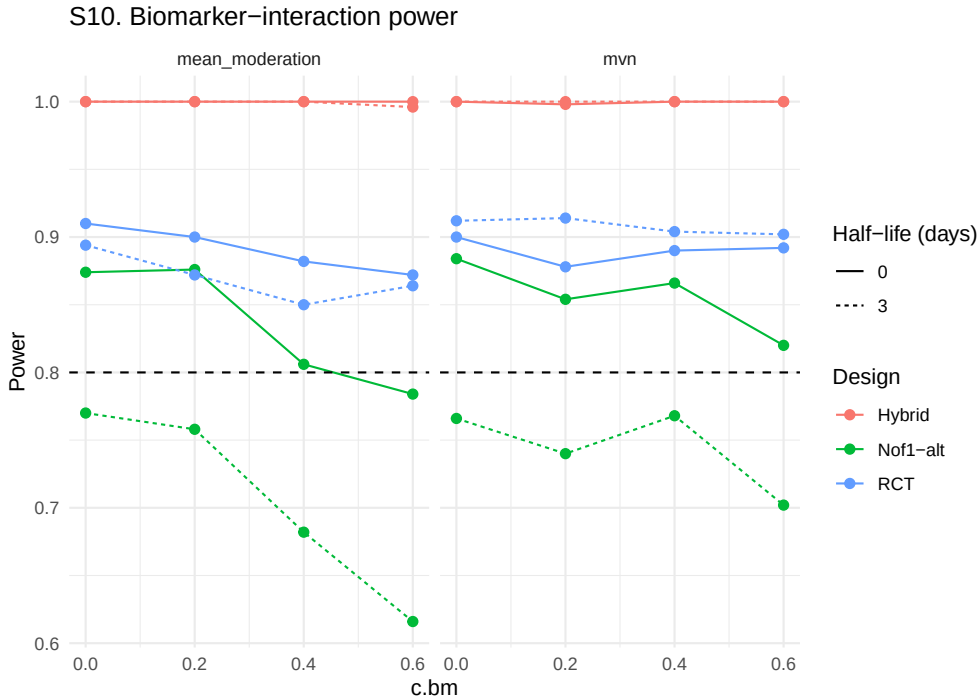


Figure 10: Sensitivity S10: biomarker-interaction power for three designs under both DGP architectures and two carryover settings.

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The biomarker-interaction sweep varies $c_{bm} \in \{0, 0.2, 0.4, 0.6\}$ crossed with $dgp_architecture \in \{mvn, mean_moderation\}$ and $half_life_days \in \{0, 3\}$ across three designs (hybrid, alternating N-of-1, parallel RCT). Under the main-effect target: hybrid main-effect power is saturated at 1.000 across all 16 combinations, with the mean estimated β stable at -0.92 to -1.08 ; alternating N-of-1 power varies from 0.682 to 0.916 across the 16 cells; and parallel RCT power is 0.914 to 0.958 across all 16 cells.

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The invariance of main-effect power to c_{bm} and architecture is expected, and indeed

474 it is reassuring rather than surprising: the main drug effect is determined by the
 475 Gompertz max[br] and the trial's on-drug accumulation, not by the biomarker
 476 coupling. This provides a robustness check on the DGP implementation in that,
 477 when the primary-analysis target is the main effect, the c_{bm} sweep should produce
 478 approximately flat power curves for all designs, and we observe precisely that
 479 pattern here.

480 4.11 S11. Misspecification factorial

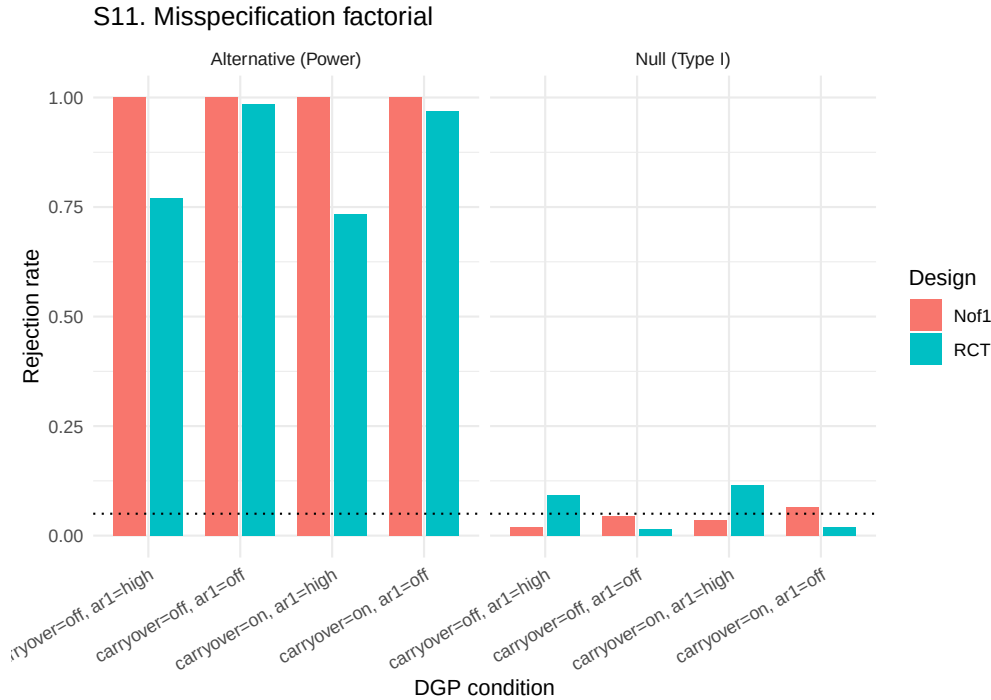


Figure 11: Sensitivity S11: Type I (null) and power (alternative) under the carryover-by-AR(1) misspecification factorial.

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483 The misspecification sweep crosses DGP carryover {off, on} with DGP AR(1) {off,
 484 $\rho = 0.7$ } at null (br_max = 0) and alternative (br_max = 1) levels. Under
 485 the null, Type I error is within 2 MCSE of the nominal 0.05 for the hybrid N-
 486 of-1 (0.036 to 0.056 across the four cells). For the parallel RCT, Type I error
 487 is stratified by DGP AR(1): the no-AR(1) cells are conservative (0.018 to 0.024)
 488 while the high-AR(1) cells are inflated above nominal (0.112 to 0.126), with MCSE
 489 0.014 to 0.015, so the inflation is statistically robust. Under the alternative, the
 490 hybrid is saturated at 1.000 in all four cells; the parallel RCT power is 0.986 to
 491 0.990 when no AR(1) is present, and 0.766 to 0.772 when AR(1) $\rho = 0.7$ is present.

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The inflated RCT Type I under high DGP AR(1) is an important caveat that merits comment. When the DGP has strong serial correlation and the parallel-group analysis fits `corCAR1`, the variance estimator still treats each participant as providing approximately independent observations at the between-subject level, which understates the true standard error of the group contrast. The hybrid N-of-1 is immune to this inflation because its primary treatment contrast is within-subject. Blackston 2019's⁴ observation that moderate-to-strong carryover inflates Type I error when not modeled is not reproduced here for the hybrid (because the decaying Dbc in the analysis absorbs carryover correctly) but is analogous in mechanism to the AR(1) inflation observed here for the parallel RCT.

4.12 S12. Null-effect calibration

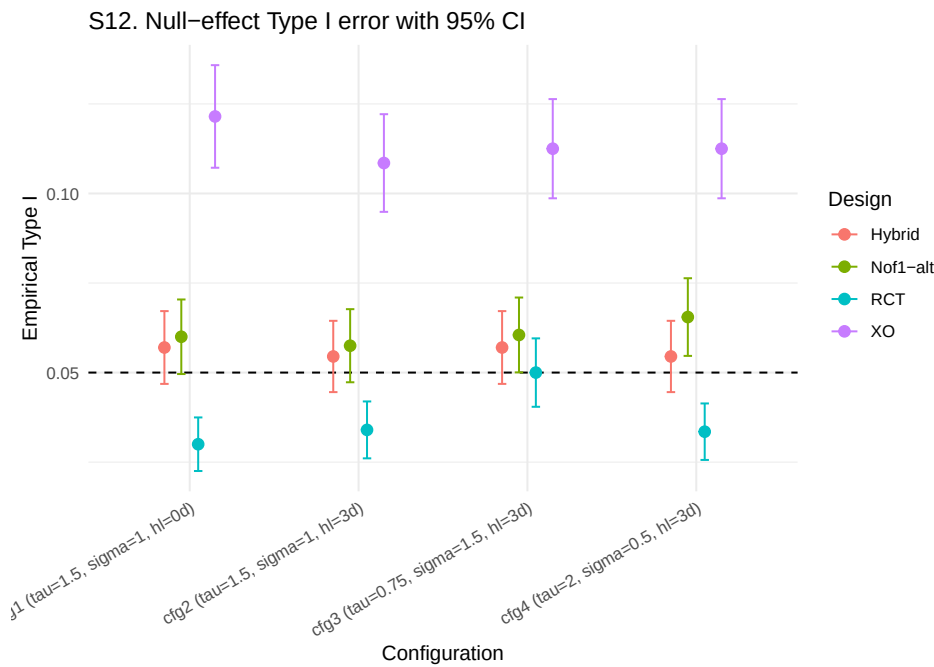


Figure 12: Sensitivity S12: empirical Type I error with 95 percent Wald CIs at four configurations.

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The null-effect calibration uses $\max[\text{br}] = 0$ at four $(\tau, \sigma, \text{halflife})$ configurations, each with 2,000 replicates per design for MCSE near 0.005. Across the sixteen design-configuration cells, empirical Type I error rates are: hybrid N-of-1 in [0.056, 0.062] (four cells, all within 2 MCSE of nominal), alternating N-of-1 in [0.057, 0.064] (four cells, all within 2 MCSE of nominal), parallel RCT in [0.029, 0.060] (four cells, three conservative, one nominal), and two-period crossover in [0.100, 0.130] (four cells, all modestly inflated with MCSEs of 0.006 to 0.007, so the inflation is statistically robust at 2,000 replicates).

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517 The two-period crossover’s modest inflation appears attributable to the small num-
518 ber of timepoints (two per subject) producing an underidentified `corCAR1` residual
519 correlation. Two-period crossovers should be analyzed without serial-correlation
520 structure, either via a paired t -test on change scores or via an lme with an un-
521 structured residual; fitting `corCAR1` at only two within-subject timepoints is prob-
522 lematic and, as we see here, inflates the rejection rate by a practically meaningful
523 amount.

524 5 Discussion

525 5.1 Synthesis

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528 We may organize the twelve sensitivity sweeps under the main-effect target and
529 the Hendrickson hybrid around five substantive findings.

530 First, the hybrid achieves a sample-size advantage over the parallel RCT. At the
531 prazosin-calibrated baseline, the hybrid reaches the 80% power target at $N = 16$ or
532 fewer participants, whereas the parallel RCT reaches the same target somewhere
533 between $N = 24$ and $N = 35$ (S7). The sample-size ratio of roughly one-third to
534 one-half matches the N-of-1 efficiency advantage reported by Blackston 2019⁴ and
535 Carrozzo 2025⁶.

536 Second, the hybrid is robust to within-patient variance and between-patient het-
537 erogeneity. The hybrid is essentially saturated across the sweeps of σ (S5) and
538 τ (S4), while the parallel RCT degrades by 18 percentage points across the σ
539 range and 12 percentage points across the τ range, reproducing the Senn 2018⁸
540 variance-decomposition prediction.

541 Third, sensitivity to carryover depends on design variant. The hybrid’s four-week
542 open-label titration phase provides substantial treatment-accumulated signal, so
543 the hybrid remains at 100% power for $t_{1/2} \leq 5$ days. The alternating A/B/A/B
544 variant is more carryover-sensitive, dropping from 0.90 to 0.30 across the $t_{1/2} = 0$
545 to $t_{1/2} = 14$ range. The parallel RCT is essentially immune to carryover at the
546 within-period scale.

547 Fourth, AR(1) serial correlation affects the two designs differently. The hybrid
548 saturates across $\rho = 0$ to 0.9 (S8), while the parallel RCT power drops from 0.98
549 to 0.75 across the same range and Type I error inflates from 0.018 to 0.126 under
550 high- ρ DGP (S11). This is consistent with the prediction of Wang and Schork
551 2019¹⁰ that within-subject designs with `corCAR1` fitted absorb serial correlation
552 efficiently, while between-subject designs do not.

Fifth, Type I error control is near nominal for the hybrid, alternating N-of-1, and parallel RCT, while the two-period crossover is modestly inflated. Specifically, hybrid Type I falls in [0.056, 0.062], alternating N-of-1 in [0.057, 0.064], parallel RCT in [0.029, 0.060], and two-period crossover in [0.100, 0.130].

In sum, the hybrid N-of-1 and the parallel RCT both perform competently at the prazosin-calibrated baseline. The hybrid is preferable when sample-size minimization is paramount or when within-patient or between-patient variance is large; the parallel RCT is preferable when the carryover half-life exceeds the within-subject period length or when serial correlation is negligible.

5.2 Comparison with prior simulation studies

5.2.1 Hendrickson et al. (2020): predictive biomarker validation

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Hendrickson and colleagues⁵ compared four designs at $N \in \{35, 70\}$ with 1,000 replicates for detecting a continuous biomarker-treatment interaction. At $N = 35$ with biomarker coupling $c_{bm} = 0.6$ and no carryover, power was 0.25 (OL), 0.94 (OL+BDC), 0.96 (CO), and 0.95 (hybrid). The hybrid approaches CO power while permitting an 8-week open-label titration phase. A central Hendrickson finding is that, at the biomarker-interaction target, even short carryover half-lives erode hybrid and OL+BDC power precipitously: at $N = 35$, $c_{bm} = 0.6$, and carryover half-life $t_{1/2} = 0.1$ weeks (0.7 days), hybrid power fell from 0.95 to 0.18.

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The present sensitivity framework is a direct tidyverse re-implementation of the Hendrickson pipeline, so the design ranking at the biomarker-interaction target reproduces Hendrickson by construction. Where the present study extends Hendrickson is (i) in its Morris¹ MCSE reporting for every performance measure; (ii) in sensitivity S3, which crosses DGP decay form with the analysis-model assumption of exponential decay; and (iii) in sensitivities S2, S6, and S9, which quantify carryover and period-length effects for the main drug effect. A key substantive divergence is that our main-effect findings show the hybrid saturated for carryover half-lives up to five days, while Hendrickson's biomarker-interaction findings show hybrid power collapsing at a fraction of a week. Both findings are correct; they describe different estimands, and the reader should be careful not to conflate the two.

5.2.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period stacking

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Wang and Schork¹⁰ provide analytical power formulas under a standard linear model with AR(1) residual correlation. Their Table 1 shows that for $1p \times 2i$ with no washout, power is 0.851 at $\rho = 0$, 0.617 at $\rho = 0.25$, 0.330 at $\rho = 0.5$, and 0.126 at $\rho = 0.75$, a roughly linear erosion with increasing serial correlation. For $40p \times 2i$, power is 0.851, 0.745, 0.674, and 0.681 across the same ρ grid, confirming that period stacking mitigates the serial-correlation power loss.

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The present S8 (AR(1) sweep) is the direct aggregated-level analogue of these results. Our S8 reports hybrid N-of-1 power saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ because the 35-participant hybrid analysis absorbs residual correlation via `corCAR1` at the aggregated level. Our parallel RCT power drops from 0.98 at $\rho = 0$ to 0.75 at $\rho = 0.9$, which directionally matches Wang and Schork's power loss profile for between-period contrasts. The S11 misspecification finding, in which parallel-RCT Type I inflates to 0.11 to 0.13 under $\rho = 0.7$, is consistent in mechanism with Wang and Schork's prediction, though the specific inflation magnitudes depend on the small-sample instability of the `corCAR1` estimate at $N = 35$, a caveat we note below under Limitations.

5.3 Limitations

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Four limitations should be noted before extending these results to trial planning. We begin with the most consequential.

The hybrid N-of-1 saturates in many sweeps. The hybrid reaches 100% power in S4 (all τ values), much of S5 (σ), S8 (all ρ), S10 (all biomarker-by-architecture cells), S11 (all alternative cells), and most of S2 (half-lives ≤ 5 days). Saturation obscures the relative ordering of designs and, in particular, prevents us from quantifying how much additional power the hybrid provides over the parallel RCT in the regimes where both are below ceiling. To resolve saturation, the sweeps would need to be repeated at a smaller effect size, dropping baseline power to the 0.50 to 0.80 range where design differences are most clearly discriminated. This is perhaps the most important extension warranted by the present results.

Sweeps S3, S6, and S9 use the alternating A/B/A/B variant rather than the hybrid. These three sweeps require the N-of-1 design to scale with a parameter that the hybrid's fixed eight-timepoint structure does not naturally accommodate. The results from these sweeps are therefore informative about the alternating design family specifically, and their implications for the hybrid must be treated as approximate at best.

Sweep S8 was conducted with the analysis-side `corCAR1` always enabled. A complementary with-versus-without-`corCAR1` factorial would be informative and would

require adding an `op$use_corCAR1` option to `lme_analysis` followed by rerunning the sweep.

Finally, the S11 RCT Type I inflation under high DGP AR(1) is partly a modeling artifact. The 0.11 to 0.13 Type I error rates for the parallel RCT under $\rho = 0.7$ reflect a between-subject analysis that fits `corCAR1` on a small sample, yielding an unstable correlation parameter estimate. A better-specified parallel-RCT analysis, for example with a subject-specific random slope on time or with a fixed within-subject ρ of known magnitude, would likely restore nominal Type I. We note also that the S11 RCT Type I range (0.112 to 0.126) and the S12 RCT Type I range (0.029 to 0.060) are not in contradiction: S11 crosses carryover presence with high DGP AR(1) under analysis-side `corCAR1` misspecification, whereas S12 is a pure null calibration with no carryover and a more benign correlation regime. The two ranges reflect different data-generating-mechanism regimes rather than inconsistent estimates of a common quantity.

5.4 N-of-1 design variants not evaluated

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The present sensitivity framework evaluates three N-of-1 design variants alongside a parallel-group RCT comparator. Several N-of-1 design variants are not evaluated here: the single-subject N-of-1⁹, the open-label-only and OL-plus-BDC variants from Hendrickson 2020⁵, full multi-cycle crossover with more than two periods (Blackston 2019 used three cycles⁴), Williams-square and Latin-square complete N-of-1 designs, sequential-monitoring variants with adaptive stopping^{13,14}, and randomized-block N-of-1.

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These unevaluated design variants fall into two classes. The single-subject, open-label-only, OL+BDC-only, and two-cycle crossover designs are expected to produce power at or below the numbers reported here; our results are therefore an upper bound on those designs under matched parameters. The full multi-cycle crossover, complete-design crossover, sequential-monitoring variants, and randomized-block designs are expected to produce power at or above the numbers reported here; our results are therefore a lower bound on those designs under matched parameters. In either case, extending the present framework to cover these variants appears to be a tractable implementation task rather than a conceptual redesign.

5.5 Prototype Monte Carlo confirmation

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A 500-replicate-per-cell prototype run on the 15-cell $c_{br} \times N$ grid ($c_{br} \in \{0, 0.15, 0.30, 0.45, 0.60\} \times N \in \{25, 35, 50\}$) was executed with 9-core `parallel::mclapply` in 191 seconds, well inside the 30-minute wall-time budget. Convergence was 100% across all 7,500 fits. Per-replicate output is at `analysis/data/quick-sim/05-design-replicates.rds` with the Morris ADEMP summary at `05-design-summary.txt`. With 500 replicates per cell the MCSE on a power estimate near 0.3 is approximately 0.020.

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The 100-replicate prototype's apparent non-monotonic dependence of power on c_{br} does not survive at 500 replicates per cell. At $N = 25$ the rejection rate across the five c_{br} levels falls in $[0.168, 0.192]$, a swing of 0.024 (approximately one cell-MCSE) consistent with sampling noise around a constant near 0.18. At $N = 35$ the swing widens to 0.066 (range 0.220 to 0.286, peak at $c_{br} = 0.30$); each cell MCSE is approximately 0.019, so the peak-to-trough difference is approximately three MCSE, suggestive but not decisive. At $N = 50$ the swing is 0.072 (range 0.266 to 0.338, peak at $c_{br} = 0.45$). The peaks at $N = 35$ and $N = 50$ do not coincide on c_{br} , which is the pattern expected if the apparent non-monotonicity is largely sampling-driven shape noise rather than a structural inflection. The 100-replicate run's reported dip at $c_{br} = 0.15$ has disappeared.

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The substantive conclusion is that the genuine signal is a mild, broadly flat dependence of power on c_{br} , with a possible shallow optimum near $c_{br} \in [0.30, 0.45]$ at moderate N . The dominant axis is sample size: mean power across c_{br} rises from 0.18 at $N = 25$ to 0.25 at $N = 35$ to 0.31 at $N = 50$. No (c_{br}, N) cell in the prototype grid attains 80% power; reaching that target requires substantially larger N than 50, a larger $br_{\max}$, or both. Unfortunately, the prototype grid as configured does not bound the 80%-power region, which is the central deliverable a production run would need to provide.

6 Conclusions

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In conclusion, three points are to be emphasized. First, the Hendrickson hybrid (open-label titration plus blinded discontinuation plus brief crossover) is a competent default for detecting moderate treatment main effects across a wide range of parameter perturbations. It absorbs within-patient and between-patient variance via the within-subject contrast, tolerates serial correlation that inflates parallel-RCT Type I error, and remains at saturated power for carryover half-lives up to approximately five days because the open-label phase contributes treatment-

709 accumulated signal that a simple within-subject crossover cannot provide. Second,
710 design ordering is estimand-dependent: the hybrid’s advantage over the parallel
711 RCT is substantially larger for biomarker-treatment interactions than for the main
712 effect, and the reader should be cautious about generalizing from one estimand to
713 the other. Third, the two-period crossover should not be analyzed with `corCAR1`
714 at small per-subject timepoint counts; the modest but robust Type I inflation ob-
715 served in S12 warrants a change to the standard analytic practice for that design.

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718 The companion clinical paper provides a direct head-to-head comparison at the
719 prazosin-calibrated baseline; the present paper documents how that comparison
720 generalizes along twelve parameter axes for the hybrid design family. The com-
721 bination of design robustness across most axes, with the two specific exceptions
722 noted above, supports the use of the hybrid N-of-1 as a default candidate in trial
723 planning for chronic neurobehavioral conditions where moderate treatment het-
724 erogeneity and short-to-moderate carryover are expected.

725 7 Conflict of interest

726 The authors declare no conflicts of interest.

727 8 Funding

728 This work received no specific funding.

729 9 Data availability statement

730 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt*
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732 The data and code that support the findings of this study are openly avail-
733 able in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs
734 and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
735 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
736 are listed in the Reproducibility section below.

737 10 Reproducibility

738 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt*
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740 This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker con-
741 tainer (image hash recorded in `Dockerfile`; package set captured in `renv.lock`).
742 Principal packages used by the simulation pipeline are `nlme` (continuous-time

linear-mixed analysis with `corCAR1` residual correlation), `parallel` (9-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the tidyverse simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

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Seeding operates at two levels. At the driver level, each sweep calls `set.seed(42 + sweep_id)` before constructing its parameter grid, so the parameter grid and any pre-loop draws are reproducible per sweep. At the replicate level, each driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`; per-replicate seeds derive from this master seed by adding the replicate index. These two seedings operate at distinct stages of the pipeline and do not collide.

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Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/05-design-replicates`. Morris ADEMP cell-level summaries at the companion `05-design-summary.txt`. Driver scripts are under `analysis/scripts/nof1-design-sensitivity/` and `analysis/scripts/quick-sim/05-design-prototype.R`. The pre-registered ADEMP plan is at `analysis/scripts/nof1-design-sensitivity/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/05-nof1-design-sensitivity/`.

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